# Xin Li

*AI Researcher | Ph.D. Student*

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## Profile

I am a Ph.D. student at NTU, working at the intersection of large language models and engineering automation with three focus areas: (1) **LLM Evaluation** — designing practical, trustworthy ways to measure LLM capabilities (DafnyComp, WirelessMathBench); (2) **LLMs for Wireless Communication** — integrating LLMs into wireless research through Project Maxwell (LACP, WirelessMathLM); (3) **LLMs for Robotics** — leveraging LLMs alongside classical perception/SLAM/navigation for embedded AI systems. Previously at MSRA, MEGVII, and Gausium Robotics, I led perception projects from prototype to deployment. I received my M.E. from Peking University and my B.E. from Northeastern University, China.

## Education

2025 – 2029 (Expected) **Nanyang Technological University**, Singapore
**Doctor of Philosophy**, Electrical and Electronic Engineering
*Advisor: Prof. Chau Yuen (IEEE Fellow)*
*Research Focus: Large Language Models for Engineering Automation*

2018 – 2021 **Peking University**, Beijing, China
**Master of Engineering**, Electronics and Communication Engineering
*Thesis: Leveraging Geometric Regularities for Robust Visual-Inertial Odometry*

2014 – 2018 **Northeastern University**, Qinhuangdao, China
**Bachelor of Engineering**, Measurement and Control Technology

## Selected Publications

*(* indicates equal contribution; see Google Scholar for complete list)*

### Large Language Models

arXiv 2025 **Local Success Does Not Compose: Benchmarking Large Language Models for Compositional Formal Verification**
Xu Xu*, **Xin Li***, Xingwei Qu, Jie Fu, Binhang Yuan
[arXiv] [Homepage]

arXiv 2025 **WirelessMathLM: Teaching Mathematical Reasoning for LLMs in Wireless Communications with Reinforcement Learning**
**Xin Li**, Mengbing Liu, Yuemei Zhu, Wenling Zhang, Li Wei, Jiancheng An, Chau Yuen
[arXiv] [Homepage]

AI4NextG @ NeurIPS 2025 **LACP: LLM Agent Communication Protocol Requires Urgent Standardization**
**Xin Li**, Mengbing Liu, Chau Yuen
[Homepage]

Technical Report 2025 **Re:Form—Reducing Human Priors in Scalable Formal Software Verification with RL in LLMs: A Preliminary Study on Dafny**
Chuanhao Yan*, Fengdi Che*, Xuhan Huang*, Xu Xu*, **Xin Li***, Yizhi Li*, Xingwei Qu*, Jingzhe Shi, Zhuangzhuang He, Chenghua Lin, Yaodong Yang, Binhang Yuan, Hang Zhao, Yu Qiao, Bowen Zhou, Jie Fu
[arXiv]

ACL 2025 Findings — **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**
**Xin Li**, Mengbing Liu, Li Wei, Jiancheng An, Merouane Debbah, Chau Yuen
[Homepage] [Code]

### Wireless Communications & Signal Processing

ML4RS@ICLR 2025 — **Onboard Terrain Classification via Stacked Intelligent Metasurface-Diffractive Deep Neural Networks**
Mengbing Liu, **Xin Li**, Jiancheng An, Chau Yuen
[Homepage]

ICASSP 2025 — **TransPathNet: A Two-Stage Framework for Indoor Radio Map Prediction**
**Xin Li**, Ran Liu, Saihua Xu, Sirajudeen Gulam Razul, Chau Yuen
[Homepage] [Code]

### Robotics & Computer Vision

IEEE RA-L 2020 — **Co-planar Parametrization for Stereo-SLAM and Visual-Inertial Odometry**
**Xin Li***, Yanyan Li*, Evin Pinar Örnek, Jinlong Lin, Federico Tombari
[arXiv] [Code]

IROS 2020 — **Leveraging Planar Regularities for Point Line Visual-Inertial Odometry**
**Xin Li***, Yijia He*, Jinlong Lin, Xiao Liu

## Work Experience

2024 – 2025 — **Nanyang Technological University**, *Research Assistant*, Singapore.
*Advisor: Prof. Chau Yuen*
Projects: Pioneered the application of LLMs to solve complex mathematical problems in wireless communications, creating the first comprehensive benchmark (WirelessMathBench) for evaluating LLMs in this domain. Developed advanced robotic perception systems integrating deep learning for real-time dense 3D reconstruction.

2022 – 2024 — **Gausium Robotics**, *SLAM Algorithm Engineer (Project Lead)*, Singapore.
Led a team of 5 engineers in developing a hierarchical visual localization system for large-scale environments, achieving 95% localization accuracy across 100,000+ sq.m. spaces. Architected real-time map update modules processing 10+ FPS on embedded systems using compact neural networks. Deployed solutions in 1000+ commercial robots worldwide.

2020 – 2021 — **Microsoft Research Asia (MSRA)**, *Research Intern*, Beijing, China.
*Mentors: Dr. Yang Liu & Dr. Yizhong Zhang*
Engineered a multi-sensor fusion system integrating RGB-D cameras and IMU data for large-scale indoor mapping. Generated high-fidelity vectorized maps of 10,000+ sq.m. supermarkets with sub-meter accuracy.

2019 – 2020 — **MEGVII Technology**, *Research Intern*, Beijing, China.
*Mentor: Dr. Yijia He*
Developed a real-time monocular Visual-Inertial Odometry system leveraging heterogeneous features, achieved semi-dense 3D mesh reconstruction at 30+ FPS. First author paper accepted at IROS 2020.

## Services

Reviewers — NeurIPS (2024, 2025), ICLR (2025, 2026), ICML (2025), AAAI (2026), AISTATS (2025), SIGGRAPH (2025), IROS (2021, 2022), ICRA (2022, 2023, 2025), IEEE RA-L, IEEE TNNLS

Organizer — AIR4D@IROS 2025 - Workshop on Advancements for Intelligent Robotics in 4D Scenes

## Awards

- 2025 Cohere Labs Catalyst Grant
- 2025 OpenAI Researcher Access Program
- 2025 PREMIA Best Student Paper Finalist
- 2025 NTU Research Scholarship (Full Ph.D. Funding)